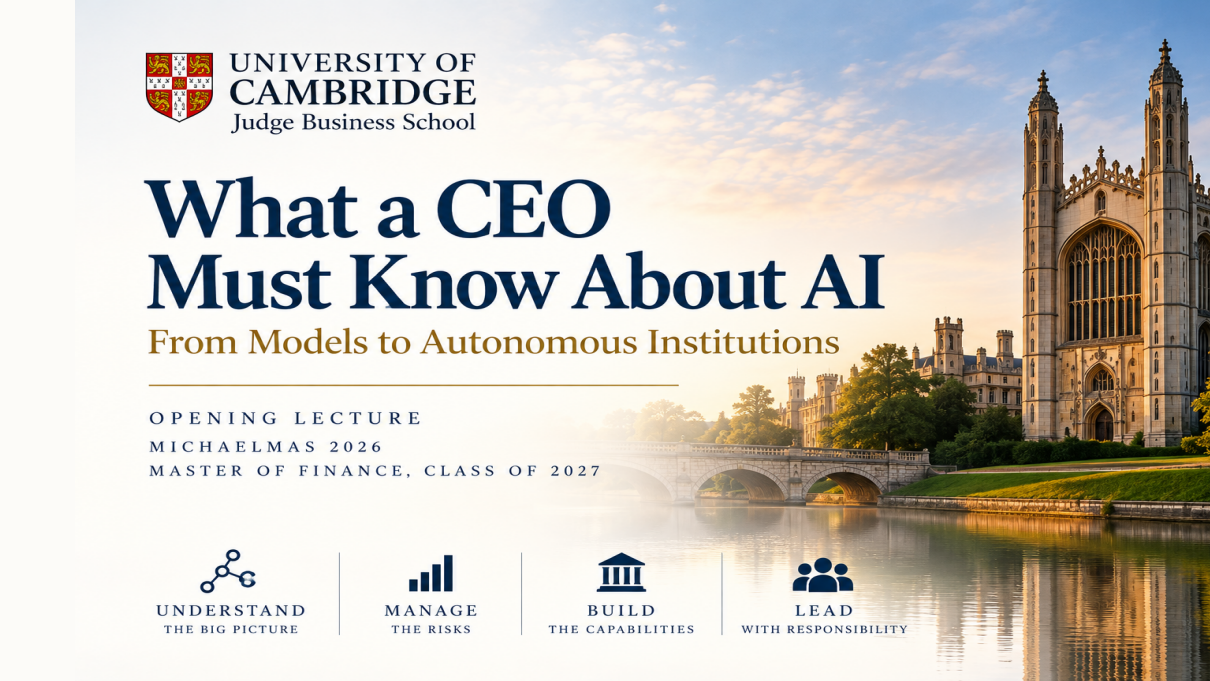
UNIVERSITY OF CAMBRIDGE
Judge Business School

What a CEO Must Know About AI

From Models to Autonomous Institutions

OPENING LECTURE
MICHAELMAS 2026
MASTER OF FINANCE, CLASS OF 2027



UNDERSTAND
THE BIG PICTURE

MANAGE
THE RISKS

BUILD
THE CAPABILITIES

LEAD
WITH RESPONSIBILITY

The course begins with purpose



01

Think as a CEO, CFO or board member: begin with the institutional decision.

02

Michaelmas: understand why AI matters to governance, strategy and risk.

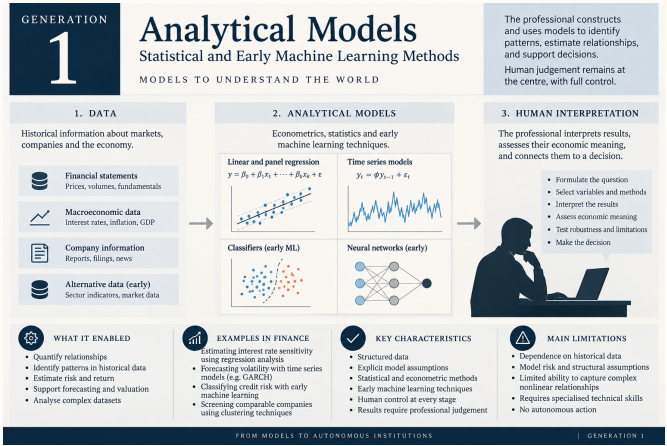
03

Easter and summer: deepen the practical mastery of tools and implementation.

04

The essential question: what should the institution become capable of doing?

Generation 1 Analytical models



01

Statistics, econometrics and early machine learning reveal patterns in data.

02

The professional chooses variables, specifies models and tests assumptions.

03

Human judgement interprets economic meaning and makes the decision.

04

Leadership responsibility: analytical validity and model limitations.

Generation 2 Conversational intelligence



01

Language becomes the interface to documents, arguments, histories and knowledge.

02

LLMs make synthesis, comparison and exploration accessible through dialogue.

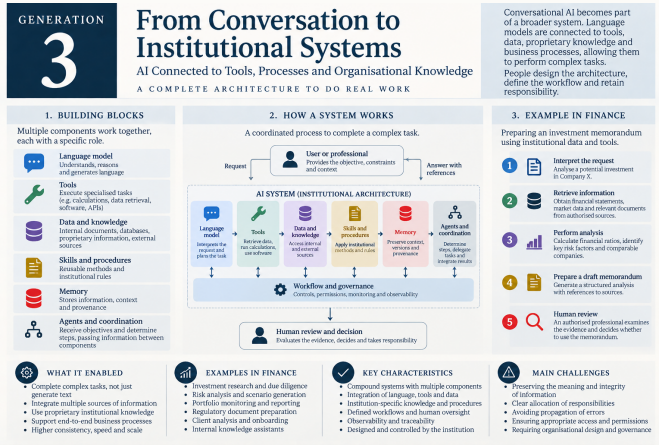
03

Fluent answers are not necessarily reliable: meaning and evidence must be checked.

04

Leadership responsibility: definitions, provenance and justified conclusions.

Generation 3 Institutional systems



01

A conversation alone cannot execute an institutional process.

02

Combine models with tools, procedures, data, memory and coordinated agents.

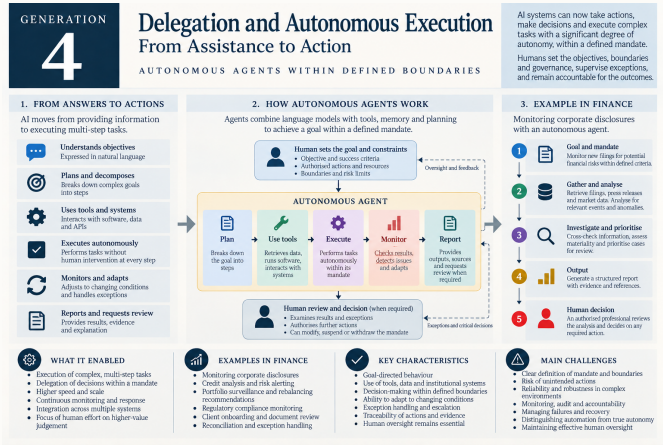
03

Humans design the workflow, approval points and operational controls.

04

Leadership responsibility: architecture, accountability and observability.

Generation 4 Bounded autonomy



01

Automation follows a prescribed path; autonomy selects actions within a mandate.

02

Agents can investigate, use tools, respond to events and escalate exceptions.

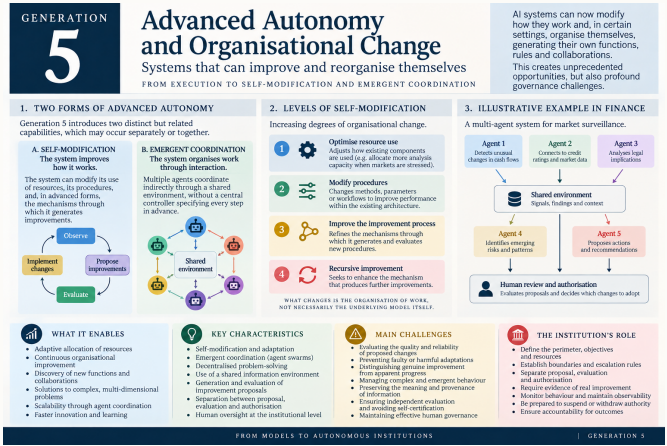
03

Technical capability never confers institutional authority.

04

Leadership responsibility: delegation limits, supervision and suspension.

Generation 5 Organisational change



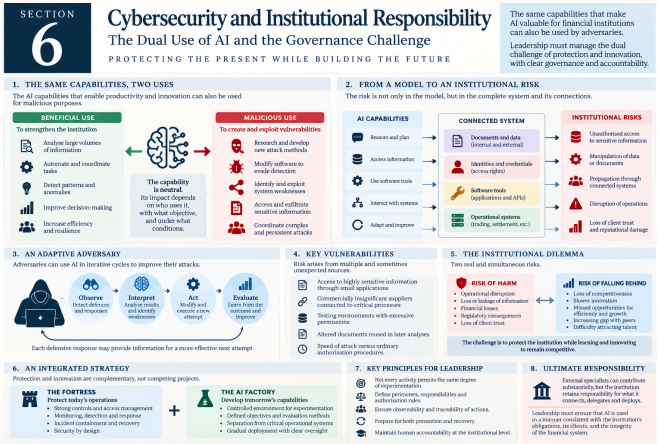
01
Self-modification changes how work is performed and improved.

02
Emergent coordination changes how agents divide and organise work.

03
Neither novelty nor agent consensus establishes genuine improvement.

04
Leadership responsibility: separate proposal, independent evaluation and approval.

Section 6 Cybersecurity and the AI dilemma



01

The same adaptive capabilities can serve institutions or adversaries.

02

Risk grows through connections to credentials, data, tools and operations.

03

Protect the present without sacrificing the ability to innovate.

04

Leadership responsibility: a secure fortress and a controlled AI factory.

Section 7 Four decisions for the CEO



01

Protect: map dependencies, contain incidents and test recovery.

02

Innovate: establish a controlled AI factory with staged deployment.

03

Invest: fund data, people, integration, testing and security—not licences alone.

04

Educate: begin with leadership; develop judgement across the institution.

The responsibilities accumulate



The task of leadership is not to adopt the most sophisticated AI. It is to choose, govern and sustain the right capabilities.